Journal of Building Engineering 54 (2022) 104634

Contents lists available at ScienceDirect

Journal of Building Engineering

journal homepage: www.elsevier.com/locate/jobe

Deterioration of basalt/carbon-based hybrid-FRP bars in a simulated pore solution of seawater sea-sand concrete

Xiangke Guo [a,b], Chuansheng Xiong [a,b,*], Zuquan Jin [a,b,**], Tao Sun [a,b]

[a] School of Civil Engineering, Qingdao University of Technology, Qingdao, 266033, PR China
[b] Engineering Research Center of Concrete Technology Under Marine Environment, Ministry of Education, Qingdao, 266033, PR China

ARTICLE INFO

Keywords:
Basalt/carbon-based hybrid-FRP bars
Seawater sea-sand concrete
Tensile strength
Deterioration

ABSTRACT

Hybrid-fibre-reinforced polymer (HFRP) bars comprising basalt and carbon fibres are expected to be more durable than BFRP bars in seawater sea-sand concrete (SWSC) environments. This study was aimed at investigating the effects of carbon-fibre content on the durability of newly developed HFRP bars conditioned in a simulated SWSC pore solution. Tensile strength was adopted as a measure of the degree of deterioration and correlated with water uptake. The deterioration mechanism was analysed by digital microscopy (DM), Fourier-transform infrared (FTIR) spectroscopy, nuclear magnetic resonance (NMR) spectroscopy, matrix digestion analysis, and inductively coupled plasma mass spectrometry (ICP-MS). The results indicated that increasing the carbon-fibre content clearly reduced the water uptake and tensile strength attenuation rate of the HFRP bars. An unambiguous correlation was established between tensile strength and water uptake. The resin swelled, hydrolysed, and plasticised after absorbing water, which initiated the deterioration of the HFRP bars. The annular basalt–carbon-fibre interfacial areas in the HFRP bars alleviated their overall deterioration but induced local degradation. The findings of this study provide insight into the development of FRP materials with improved durability.

1. Introduction

Seawater sea-sand concrete (SWSC) has opened a new avenue for the production of concrete using locally sourced, original sea-sand and seawater in marine construction. This strategy can offer compelling environmental and economic advantages such as alleviating the excessive extraction of river sand that destroys river ecosystems, reducing the competition for freshwater use between production and living, and saving the long-distance transportation costs of construction materials [1]. However, the high concentration of chloride ions in SWSC poses a serious threat to steel bars [2,3]. Fibre-reinforced polymer (FRP) bars are ideal substitutes for steel bars to strengthen SWSC owing to their superior corrosion resistance to chloride attack [4]. Additionally, structures designed using FRP bars exhibit decent characteristics such as long-term durability and non-magnetism. However, as internal reinforcements, FRP bars are continuously exposed to a strong alkaline environment that can degrade the material properties and their bonding to concrete.

FRP bars degrade in strongly alkaline environments primarily because of the deterioration of the resin and fibre–resin interface. The resin degradation is primarily caused by hydrolysis via abundant OH^- ions, which also increases the void content of FRP materials, thereby weakening the fibre–matrix interface [5]. The deterioration manifests as a loss of mechanical properties. Lu et al. [6] found

* Corresponding author. School of Civil Engineering, Qingdao University of Technology, Qingdao, 266033, PR China.
** Corresponding author. School of Civil Engineering, Qingdao University of Technology, Qingdao, 266033, PR China.
E-mail addresses: xiongcs@qut.edu.cn (C. Xiong), jinzuquan@126.com (Z. Jin).

https://doi.org/10.1016/j.jobe.2022.104634
Received 27 February 2022; Received in revised form 19 April 2022; Accepted 7 May 2022
Available online 11 May 2022

that conditioning FRP bars comprising basalt (BFRP), glass (GFRP), and carbon (CFRP) in alkaline solutions (pH = 12.9) reduced their residual tensile strength (TS) compared to those of bars in water and seawater. Yi et al. [5] comparatively investigated the durability of BFRP bars in SWSC solutions with different alkalinities (pH = 10.1, 12.3, and 13.2). Increasing the alkalinity increased moisture uptake and degraded the mechanical properties, including interlaminar shear and transverse shear strength. Moreover, Lu et al. [7] noted the importance of alkalinity in enabling the damage of BFRP materials. Similar results have been reported on GFRP bars subjected to sustained stress in concrete environments, as reported by D'Antino and Pisani [8]. Moreover, the instability of FRP bars in alkaline environments significantly hinders their popularisation and application in concrete structures.

The water or OH^- contents of alkaline solutions also induce fibre degradation. The main component of basalt/glass fibres—SiO_2—induces the alkali–silica reaction by absorbing water and swelling in alkaline solutions, which decreases the effective diameter and strength of the fibres [9]. In contrast, the tortuous and long diffusion paths of water molecules and detrimental ions in CFRP bars due to the narrow spacing of the carbon fibres reduces the degradation rates of the resin and fibre [10]. Additionally, the excellent durability and mechanical properties of carbon fibre enable CFRP bars to exhibit remarkable durability in alkaline environments. Nevertheless, the high costs of CFRP bars hinder their use in concrete. Therefore, carbon fibre can ideally be used to replace a portion of basalt fibre (glass fibre) to form basalt (glass)–carbon hybrid FRP (HFRP) bars, which are anticipated to exhibit superior mechanical properties and longer service life than those of BFRP or GFRP bars (Fig. 1).

Glass–carbon HFRP bars have attracted recent scientific attention, and have been typically configured in three forms by varying the arrangement of CFRPs (Fig. 2). Compared with GFRP bars, glass–carbon HFRP bars retain their original interlaminar shear strength but show improved corrosion resistance owing to reduced water uptake and OH^- diffusion rates [10–12]. BFRP bars are increasing in popularity and are being gradually applied in civil-engineering arenas. Generally, BFRP bars are environmentally friendly and have similar or higher stiffness and strength than those of GFRP bars; moreover, it is inexpensive compared to CFRP bars [1]. Nevertheless, the durability of basalt–carbon HFRP bars has not been thoroughly investigated. It is worth noting that the 'HFRP bars' notation used in this paper corresponds to the basalt–carbon HFRP bars.

In our previous study [14], the arrangement of carbon fibres in HFRP bars was found to be insignificant in influencing their mechanical behaviour. Therefore, to comprehensively investigate the effects of carbon-fibre hybridisation content on the durability of HFRP bars in a simulated SWSC pore solution based on tensile strength, HFRP bars were designed in this study by wrapping a layer of BFRP around CFRP and incorporating the CFRP into the core of the BFRP. Carbon-fibre volume fractions of 0%, 10%, and 25% (denoted as BFRP, H1FPR, and H2FPR, respectively) were used in the HFRP bars. Furthermore, the deterioration of the HFRP bars in different simulated SWSC environments was analysed by water uptake analysis, digital microscopy (DM), Fourier-transform infrared (FTIR) spectroscopy, nuclear magnetic resonance (NMR) spectroscopy, matrix digestion analysis, and inductively coupled plasma mass spectrometry (ICP-MS).

2. Experimental

2.1. Materials

HFRP bars with a nominal diameter of 10 mm were used. Small bars were selected to achieve faster water uptake and obtain more conservative durability results than those of larger specimens [9]. Bisphenol-A resin and methyltetrahydro-phthalic anhydride hardener (5:4 by weight) were used as the matrix materials. The detailed parameters related to the cured epoxy matrix and fibres (Fig. 3) are listed in Table 1. A fibre volume fraction of ~65% by mass was used. The pre-arranged carbon and basalt fibres were impregnated into the resin matrix and fabricated by pultrusion (Fig. 4). Additionally, the surfaces of the HFRP bars were ribbed with a nylon laminate during their production (Fig. 5) to ensure bonding with concrete. The average inner and outer diameters of the HFRP bars were estimated to be 9.94 ± 0.1 and 10.92 ± 0.1 mm, respectively. The average diameter (10 mm) was adopted to calculate the cross-sectional area of the bars.

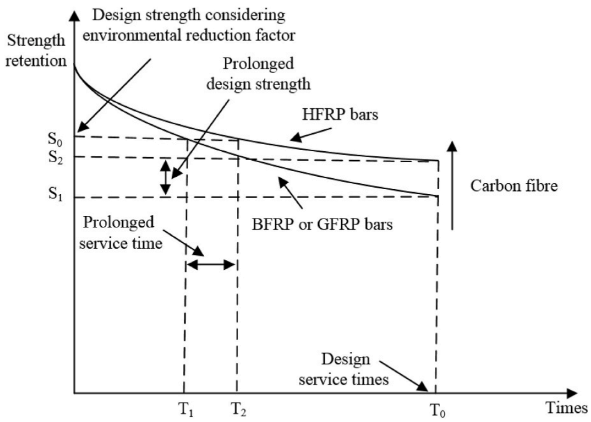


Fig. 1. Benefits of HFRP bars.

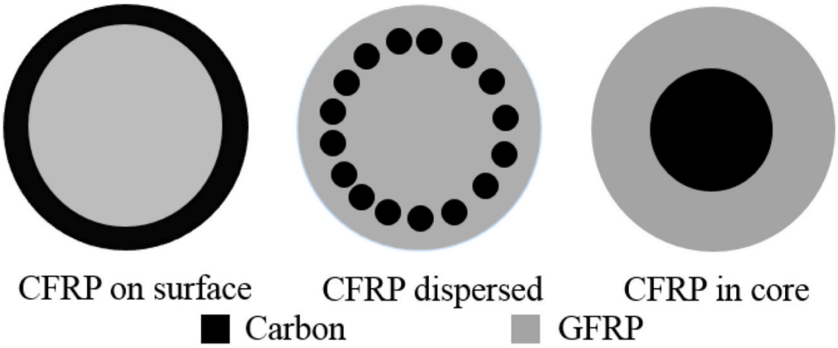


Fig. 2. Cross sections of HFRP bars [10–13].

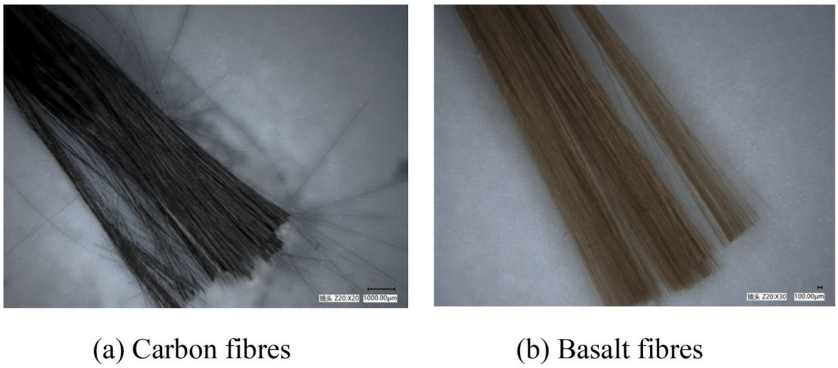


Fig. 3. Physically drawn fibres.

Table 1
Properties of fibres and cured epoxy matrix.

Sample component	Density (g/cm^3)	Diameter (μm)	Tensile strength (MPa)	Tensile modulus (GPa)
Carbon fibre	1.8	8	5610	295
Basalt fibre	2.65	15.8	3800–4810	93.1–110
Cured epoxy matrix	/	/	55.2	3.62

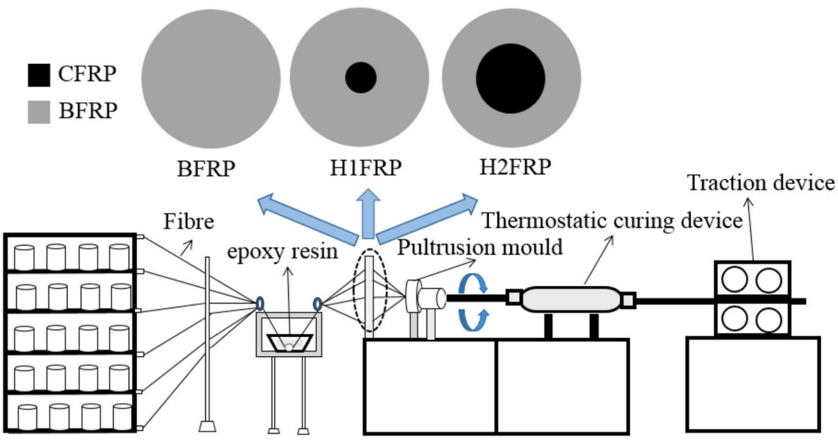


Fig. 4. Schematic of the fabrication of HFRP bars.

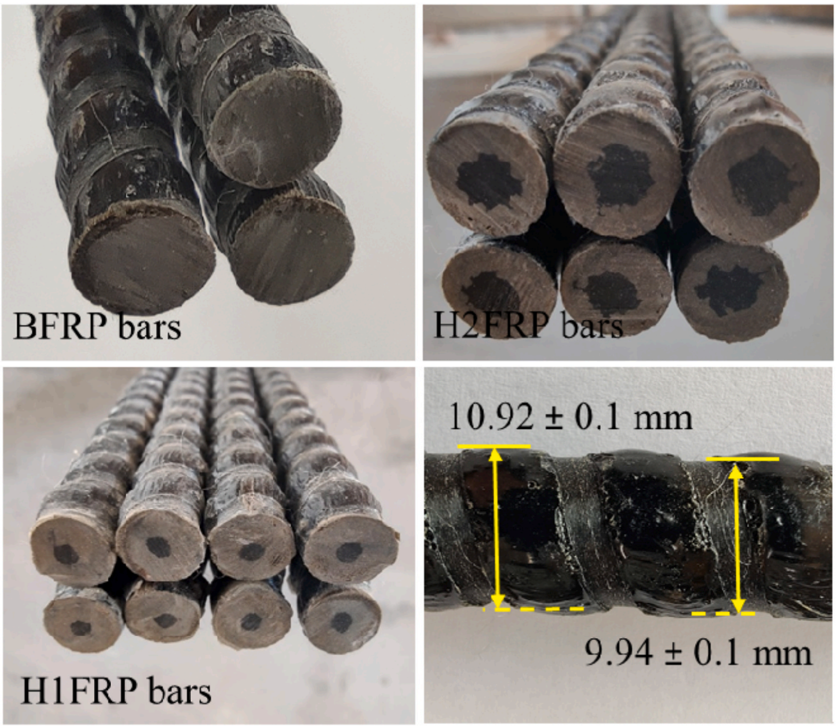


Fig. 5. Photographs of the HFRP bars.

2.2. Conditioning

An alkaline SWSC solution was used to simulate an SWSC pore solution for the HFRP bars, based on the protocol described by Yi et al. [5]. The composition of the solution is described in Table 2. The seawater was derived from the shallow sea area of Qingdao, Shandong Province, China. The concentrations of ions in the seawater are listed in Table 3. Three conditioning temperatures (laboratory temperature (LT), 40 °C, and 55 °C) and durations (30, 60, and 90 d) were adopted to determine their effects on the durability of the HFRP bars. These temperatures accelerate the aging of the FRP bars but do not alter their degradation mechanism after conditioning [15]. The LT was monitored using an intelligent temperature-humidity meter during the exposure tests, and the average temperature was recorded to be ~21 °C.

2.3. Water uptake tests

The moisture uptake of the HFRP bars was measured according to the ASTM D570 protocol [16]. For each exposure condition, dry and pre-weighed (w_0) specimens were conditioned in the previously described environment. Five 50-mm-long samples were employed for each conditioning test, and their ends were sealed with silicone to enable analysis of water uptake in the radial direction. The samples were removed and surface dried with a paper towel after specified durations and subsequently weighed (w_t) using a digital balance with a minimum limit of 0.001 g. The moisture uptake of the FRP specimens after conditioning (%, by mass) was estimated using Eq. (1).

$$W_t = \frac{w_t - w_0}{w_0} \times 100\% \tag{1}$$

2.4. Tensile tests

The HFRP bars were prepared, exposed, and subjected to tensile testing. The long-term durability of the bars was evaluated based on their tensile properties in accordance with the ACI 440.3R-12 protocol [17]. The production of the tensile specimens is described elsewhere [14]. A uniaxial tensile load was applied to the test specimens at a rate of 2 mm/min in the displacement control mode. The applied load and displacement results were recorded using a data acquisition system, and the strain was measured using an extensometer. It is worth noting that the yield stage did not appear when the FRP bars failed. Therefore, the extensometer was manually

Table 2
Simulation of an SWSC pore solution.

NaOH (g/L)	KOH (g/L)	$Ca(OH)_2$ (g/L)	pH
2.4	19.6	2	13.2

Table 3
Concentration of ions in the seawater (mg/L).

SO_4^{2-}	Mg^{2+}	Cl^-	Ca^{2+}
2388	1401	14501	344

removed when the load exceeded 60% of the ultimate load to prevent damage. A schematic and photograph of the tensile test setup are shown in Fig. 6.

2.5. Digital microscopy (DM)

DM (VHX-1000C) observations and image analysis were conducted on selected HFRP bars to assess the effects of duration and temperature on morphological changes. The micrographs were expected to reveal the deterioration mechanism of the HFRPs. Moreover, BFRP specimens with damaged fibres, polymer matrix, and/or fibre–matrix interface were used for comparison.

2.6. FTIR spectroscopy

The chemical nature of unexposed HFRP specimens was examined using a BRUKER FTIR apparatus with an attenuated total reflectance (ATR) setup. Samples for FTIR analysis were also acquired from the surfaces of specimens exposed to 55 °C for 90 d. To facilitate analysis of the HFRP bars surfaces, the exposed bars were ultrasonically cleaned and naturally dried. Sixty-four scans were acquired in the wavenumber range of 500–4000 cm^{-1} at a resolution of 4 cm^{-1}.

2.7. NMR spectroscopy

The chemical compositions of the resin matrix after 90 d of exposure at 55 °C were determined by ^1H NMR and ^{13}C NMR spectroscopies. After FTIR analysis, the surfaces of the conditioned samples were pulverised, dried at 40 °C for 72 h, and analysed using a 400 MHz Vaian DLG400 NMR setup. The system was attached to a magnet operating at 400 MHz with a beta barium borate positive-phase observation broadband probe. Spectra were obtained at 50 °C, and samples were prepared by dissolving 50 mg of the powdered sample in deuterated dimethyl sulfoxide-d_6 (DMSO-d_6) in an NMR tube. DMSO-d_6 was used as a reference at 2.54 and 40.45 ppm in the ^1H NMR and ^{13}C NMR analyses, respectively [18].

2.8. Matrix digestion analysis

The matrix contents of the exposed and unexposed HFRP bars were determined according to the ASTM D3171 protocol [19]. Typically, nitric acid can dissolve epoxy resin without affecting the fibres. The procedure was repeated three times for each type of sample in the present study. First, the original mass (m_0) of each specimen was determined, and the specimen was added to ~50 mL of 70% aqueous nitric acid. The obtained mixture was placed in a water bath at 80 °C for 6 h. Subsequently, the fibre mass (m_f) was

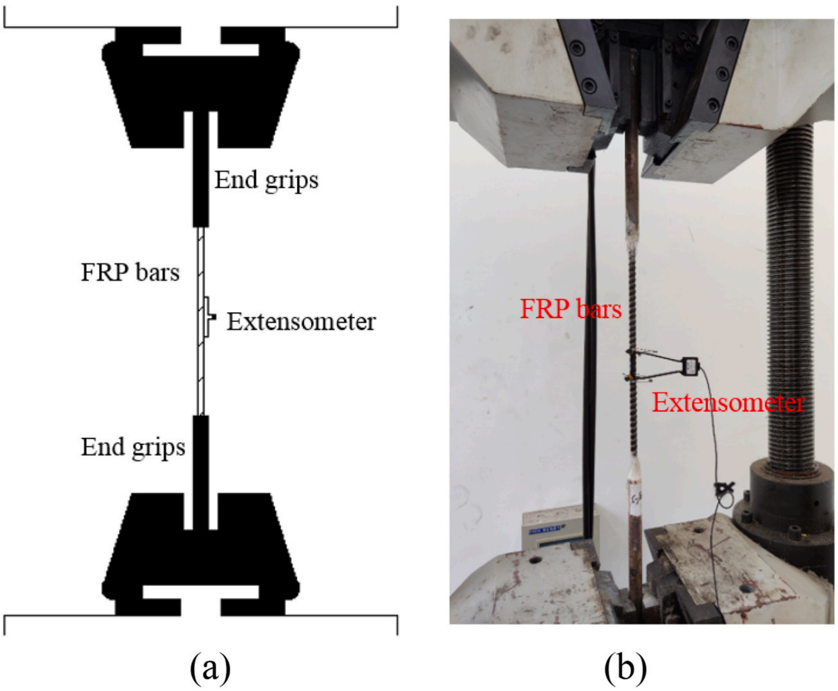


Fig. 6. (a) Schematic and (b) photograph of the tensile testing setup.

determined after residual material filtration, washing in distilled water, and natural drying. The matrix retention (M_t) was calculated as follows:

$$M_t = \frac{m_0 - m_f}{m_0} \times 100\% \tag{2}$$

2.9. Inductively coupled plasma mass spectrometry (ICP-MS)

Fibre leaching, which qualitatively represents the degree of HFRP deterioration under different conditioning temperatures, was characterised by ICP-MS. Following the protocol established by Yi et al. [5], the residual solution after conditioning was collected in a centrifuge tube, and a 1 mL suspension was extracted into a Teflon beaker. Subsequently, nitric acid (68%, 1 mL), hydrochloric acid (37%, 1 mL), and hydrofluoric acid (25%, 1 mL) were added to the digestion beaker. An electrothermal digester was used to digest the specimens, and a test tube with a constant volume (10 mL) was used to collect the filtered solution. The concentrations of the leached elements in the fibres were finally determined by ICP-MS spectrometry.

3. Results and discussion

3.1. Water uptake

Water uptake significantly affects the mechanical properties and durability of FRP materials. Yi et al. [5] found strong evidence of the degradation rate of BFRP bars being related to the rate of water-based liquid uptake. FRP bars in alkaline water-based solutions absorb water molecules and OH^- ions, leading to hydrolysis and plasticisation of the resin and deterioration of the fibres and fibre–resin interface, thereby reducing their tensile strength [20].

The protocol reported by Manalo et al. [9] and Pan and Yan [10] was adopted in this study to acquire quantitative absorption and diffusion data of the conditioned HFRP bars based on water uptake. Fig. 7 shows the water uptake data of the conditioned HFRP samples plotted against the square root of time (in seconds) at different temperatures. The water uptake of the bars was expected to be higher at the surfaces rather than their ends, which were sealed. Table 4 lists the diffusivity coefficient (D_r) and saturated water

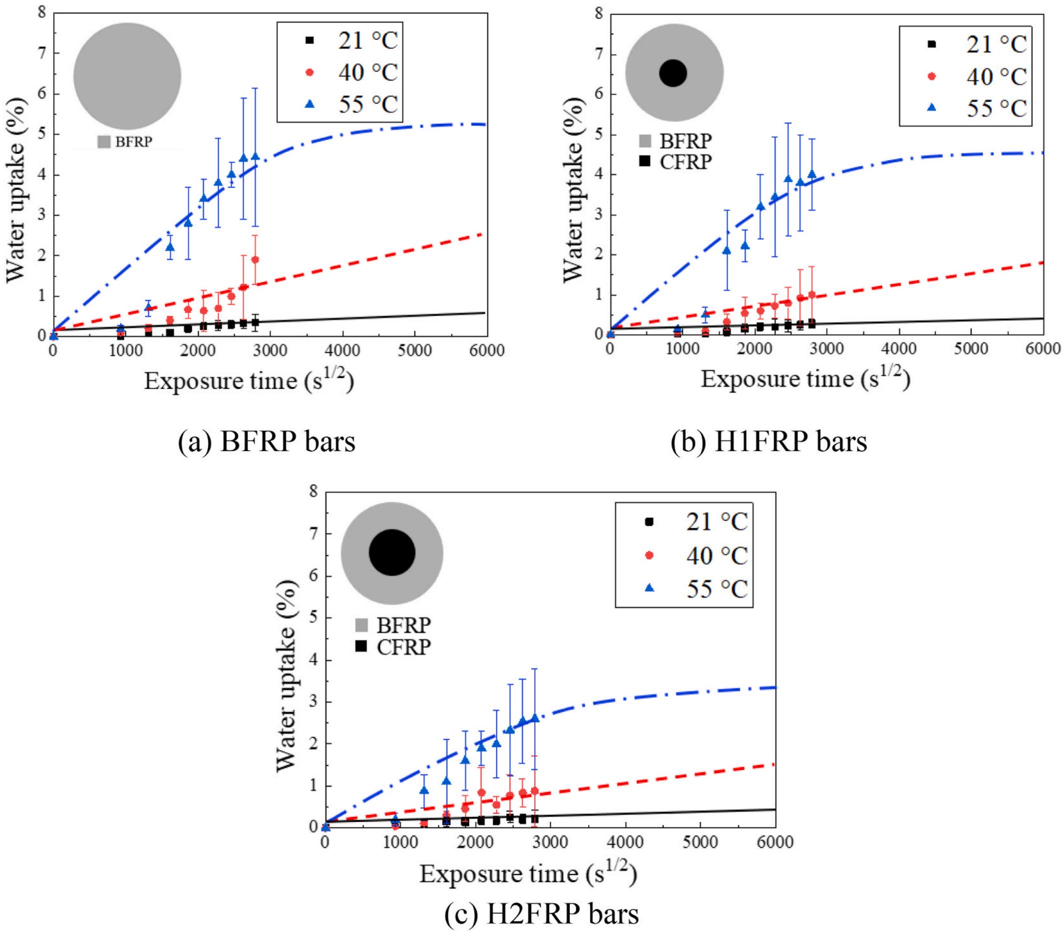


Fig. 7. Moisture uptake of the HFRP specimens.

Table 4
D_r and W_∞ data of the HFRP specimens.

FRP specimen	Temperature (°C)	D_r (10^{-13} m^2/s)	W_∞ (%)
BFRP	21	0.09	5.18
	40	1.89	
	55	5.04	
H1FRP	21	0.07	4.51
	40	1.72	
	55	4.77	
H2FRP	21	0.03	3.49
	40	1.49	
	55	3.98	

absorption (W_∞) data of the conditioned HFRP specimens, which were calculated using theoretical Fickian curves (Eq. (3)).

$$\frac{W_t}{W_\infty} = 1 - \sum_{n=1}^{\infty} \frac{4}{r^2\alpha_n^2} \exp\left(-D_r\alpha_n^2 t\right) \tag{3}$$

where W_t (%) represents the absorbed moisture, W_∞ (%) is the saturation level of the absorbed water, D_r (mm^2/s) is the radial diffusion coefficient, t (s) is the exposure time, r (mm) is the radius of the HFRPs, and α_n is the nth root of the zeroth-order Bessel function. The cylindrical geometry of the HFRP samples necessitated the inclusion of a Bessel function in Eq. (3).

The water uptake data were scattered owing to the relatively small variety in mass. Overall, the water uptake of the HFRP specimens followed the theoretical Fickian law, which typically classifies these processes into two stages [11]. As shown in Fig. 7, the water uptake of all specimens initially increased in a quasilinear manner and gradually saturated at 55 °C but not at 21 and 40 °C, indicating the significance of the exposure temperature in water absorption. The chemical structure of the epoxy resin and the fibre volume fraction determine the W_∞ values of FRP materials [21]. High temperatures accelerated structural relaxation of the epoxy resin, which increased its rate of absorption of water molecules. Similarly, the D_r values also increased with increasing exposure temperature. Additionally, the BFRP bars had the highest D_r and W_∞ values (5.04^{-13} m^2/s and 5.18%, respectively; Table 4), followed by the H1FRP ($W_\infty = 4.51\%$, $D_r = 4.77$ m^2/s) and H2FRP bars ($W_\infty = 3.49\%$, $D_r = 3.98$ m^2/s). The low D_r and W_∞ values of the H1FRP and H2FRP bars were due to the low absorption and diffusion of moisture in CFRP [22], which can mitigate the OH^--induced breakage of the fibre structures, depolymerisation of the resin, and debonding of the fibre–resin interface.

The W_∞ value of the BFRP bars (5.18%) was slightly lower and significantly higher than those obtained by Yi et al. [5] (5.47%) and Wang et al. [22] (15.82%), respectively. This was presumably because the exposed solution used by Yi et al. [5] was simulated seawater rather than natural seawater; the former contains larger or a greater number of ions such as salt ions than those in the latter. With respect to the result obtained by Wang et al. [22], the small diameter of the specimens and their unsealed ends led to the significant difference.

3.2. Tensile properties

The tensile properties of the HFRP specimens are listed in Tables 5–7, and those of reference HFRP specimens are depicted in Fig. 8. Tensile strength, elastic modulus, and ultimate tensile strain were used as evaluation parameters [14]. All parameters except for the ultimate tensile strain considerably increased with increasing carbon-fibre hybridisation from 0% to 10% and 25%. The tensile strength and elastic modulus increased from 1216.18 to 1705.99 MPa and 61.17 to 85.01 GPa, respectively, with increasing carbon-fibre hybridisation from 0% to 25%. This was due to the higher elastic modulus and tensile strength of carbon fibre than those of basalt fibre.

Table 5
Tensile properties of BFRP.

Temperature (°C)	Duration (d)	Tensile strength (TS; MPa)		Elastic modulus (GPa)		Ultimate tensile strain (%)	
		Average value	COV (%)	Average value	COV (%)	Average value	COV (%)
21	Ref.	1216.18	2.35	61.17	5.13	2.01	1.99
	30	1225.23	1.41	61.34	1.81	1.95	2.14
	60	1187.39	1.50	58.09	2.10	2.08	0.97
	90	1163.78	3.39	57.29	4.11	2.28	3.18
40	30	1168.17	3.41	61.93	1.98	2.23	1.72
	60	1076.43	1.40	55.47	3.18	2.44	1.88
	90	980.05	7.18	56.43	0.81	2.09	5.17
55	30	1082.8	2.12	51.63	11.87	1.62	0.97
	60	597.68	2.41	58.09	8.77	1.15	4.79
	90	318.68	3.60	51.66	3.81	1.76	15.17

COV: coefficient of variation.

Table 6
Tensile properties of H1FRP.

Temperature (°C)	Duration (d)	TS (MPa)		Elastic modulus (GPa)		Ultimate tensile strain (%)	
		Average value	COV (%)	Average value	COV (%)	Average value	COV (%)
21	Ref.	1385.43	2.53	72.11	6.33	2.44	2.87
	30	1393.55	2.10	73.48	4.04	2.49	8.48
	60	1334.61	1.92	70.94	2.65	2.46	3.78
	90	1270.06	1.70	70.48	0.97	2.09	0.90
40	30	1324.56	2.73	74.77	1.77	2.48	2.17
	60	1066.04	9.35	68.65	9.36	1.78	9.09
	90	1054.89	3.60	68.25	5.44	1.62	5.52
55	30	1157.63	3.81	70.98	3.12	1.71	0.88
	60	847.76	2.10	67.09	0.31	1.79	3.17
	90	509.55	3.32	68.23	3.89	1.21	2.90

Table 7
Tensile properties of H2FRP.

Temperature (°C)	Duration (d)	TS (MPa)		Elastic modulus (GPa)		Ultimate tensile strain (%)	
		Average value	COV (%)	Average value	COV (%)	Average value	COV (%)
21	Ref.	1705.99	3.28	85.01	2.97	1.98	7.07
	30	1644.10	2.10	83.07	1.54	1.89	3.05
	60	1693.50	2.69	80.70	0.97	1.94	1.15
	90	1663.69	3.81	82.55	3.17	1.97	0.85
40	30	1672.72	1.73	81.75	0.66	2.18	3.17
	60	1552.29	3.20	82.61	0.98	2.23	1.87
	90	1420.95	3.74	80.36	3.12	1.87	9.50
55	30	1570.03	4.58	85.49	6.07	1.81	1.17
	60	1195.14	3.80	85.59	3.02	1.76	3.14
	90	942.68	7.90	80.47	11.79	1.23	1.08

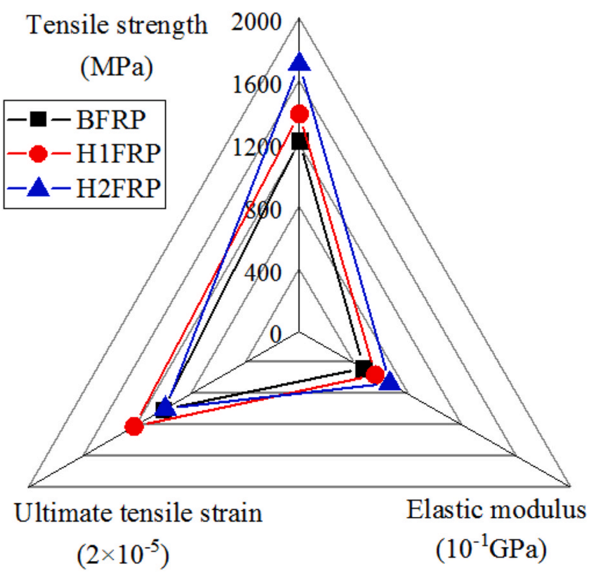


Fig. 8. Tensile properties of reference HFRP specimens.

3.2.1. *Tensile strength (TS)*

The long-term properties of the conditioned HFRP bars were characterised based on their TS. In contrast to elastic modulus and ultimate tensile strain, TS is typically used to evaluate the durability of samples with a more fibre-dominated behaviour [9,14,23], and as a sensitive parameter in structural design standards, such as the ASCE (United States) [24], EUR27666 (Europe) [25], and CECS (China) [26].

Tables 5–7 also list the coefficients of variation (COVs) of the TS of the HFRP specimens. The low COVs of the unconditioned HFRP bars (COV_{max} = 3.28%) indicate that all investigated bars were acquired from the same production lot, and the manufacturing process had little effect on their performance. However, as reported by Feng et al. [27], non-uniform fibre–resin distributions have inherent initial defects for FRPs owing to the limitations of pultrusion. The estimated COVs of the conditioned bars are slightly higher than those obtained by Rifai et al. [28] but consistent with those determined by Wang et al. [23]. The high COVs were due to the exposure of the bare FRP bars to the high-alkalinity solution, which enhanced the non-uniform moisture absorption and deterioration of the bars.

Residual tensile strength (R, %) is generally employed to quantify the effects of exposure temperature and time on the TS, and is defined as follows:

$$R = [(\sigma_0 - \sigma_t) / \sigma_0] \times 100\% \tag{4}$$

where σ_0 and σ_t are the TSs (MPa) of the unconditioned and conditioned FRP samples, respectively. Fig. 9 shows the R values of the tested HFRP bars. The following conclusions can be drawn:

- The R values of BFRP, H1FRP, and H2FRP exposed to a simulated SWSC solution at 21 °C for 90 d were determined to be 95.69%, 91.67%, and 97.52%, respectively, and the corresponding TSs were 1163.78, 1270.06, and 1663.69 MPa, respectively.
- The TSs of the BFRP, H1FRP, and H2FR bars exposed to 40 °C for 90 d were reduced by 19.42%, 23.86%, and 16.71%, respectively.
- For samples that were exposed to 55 °C for three months, a loss in TS of ~45% was recorded for H2FRP; the corresponding TS was 942.68 MPa. However, the TSs of the BFRP and H1FRP bars decreased from 1216.18 to 318.68 MPa and 1216.18 to 509.55 MPa, respectively (R = 26.2% and 36.78%, respectively).

These experimental results clearly indicate that the TS of the HFRP bars decreased with increasing duration because of the increase in water absorption by the bars with increasing exposure time. Essentially, the bars absorbed water molecules and abundant, free OH^- ions from the alkaline solution, and the resin matrix swelled, formed microcracks, and was hydrolysed and plasticised, which diminished the fibre–resin bonding force. This led to fibre–matrix interfacial debonding, delamination, and fibre etching. The rapidly absorbed and diffused water molecules and OH^- ions aggravated the deterioration of the fibres, resin, and fibre–resin interface in the HFRP bars with increasing temperature. Therefore, high temperatures such as 55 °C had an immediate effect on the degradation of the bars.

Additionally, the H2FRP bars had a higher R value than that of the BFRP specimens, especially for samples with long-term and high-temperature exposure, presumably because of the lower D_r and W_∞ values of the H2FRP bars with a relatively high carbon-fibre content. However, the BFRP bars exhibited a higher R value than that of the H1FRP bars under all investigated conditions, except for 90 d of exposure at 55 °C. A close examination of the basalt–carbon-fibre interfacial region (Fig. 10) revealed slight intermingling of the carbon fibres with the larger basalt fibres, indicating close packing. The space between the larger basalt fibres was occupied by the smaller carbon fibres. Therefore, the route for water diffusion changed from the wide spacing of the basalt fibres to the narrow spacing of the basalt–carbon-fibre interfacial region. The close packing in this interfacial region enabled an effective moisture-barrier characteristic that impeded the diffusion of water molecules and gathered OH^- ions, resulting in local deterioration of the annular BFRP in the outer layer of the interfacial region. Therefore, the R value of the H1FRPs was lower than that of the BFRP bars in the earlier stage of the conditioning but higher than that in the later stage owing to the excellent corrosion resistance of carbon fibre. Furthermore, the annular basalt–carbon-fibre interfacial region in the HFRP bars was weak, which should be focused on in future research on durability.

Fig. 11 shows TS–water-uptake data and their fitting curves. The fitting coefficients of all the HFRP data fits (R^2) were greater than 0.90, indicating a close correlation between the water uptake and TS of the HFRPs. Moreover, this corroborates the significance of water uptake in the deterioration of HFRP samples. Furthermore, this proves the effectiveness of the strategy of replacing basalt fibres with carbon fibres to form basalt–carbon HFRP bars for enhancing the durability of FRP bars, possibly owing to the low water uptake.

Fig. 12 shows the residual tensile strength (R) values of the HFRP specimens and corresponding data from previous studies conducted in similar environments. Considering the limited research on bare BFRP bars exposed to simulated SWSC solutions, studies performed on these bars using other alkaline solutions were also included. Moreover, as reported by Guo et al. [14], Yi et al. [5], and Liu et al. [20], the durability of FRP materials can be sufficiently characterised when bars are exposed to water-based solutions at

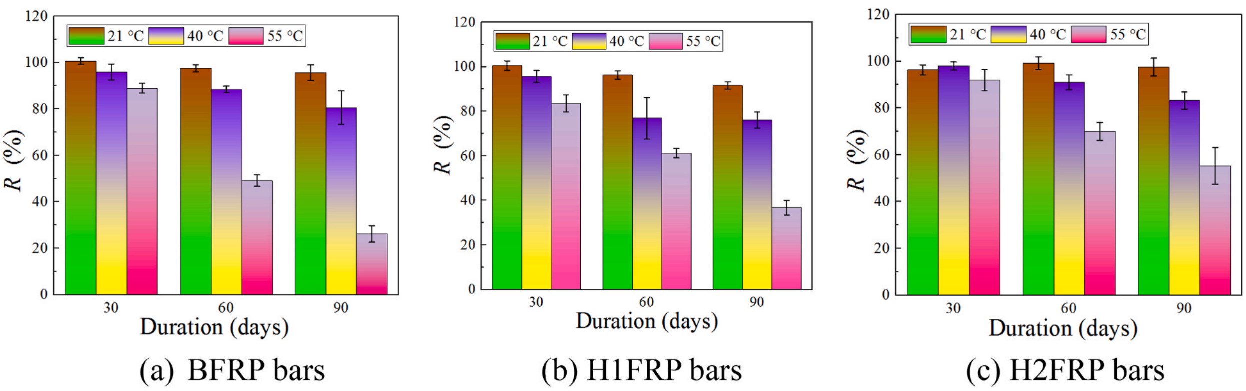


Fig. 9. Residual tensile strength of conditioned HFRP specimens.

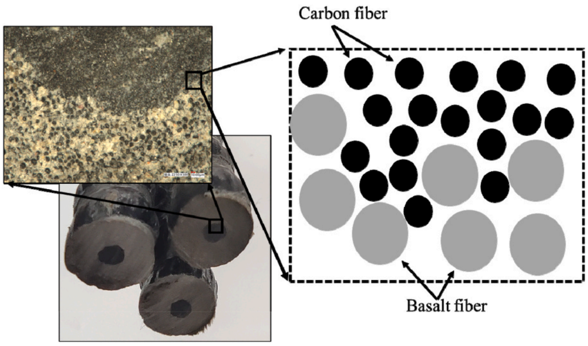


Fig. 10. Schematic of basalt–carbon-fibre interfacial region.

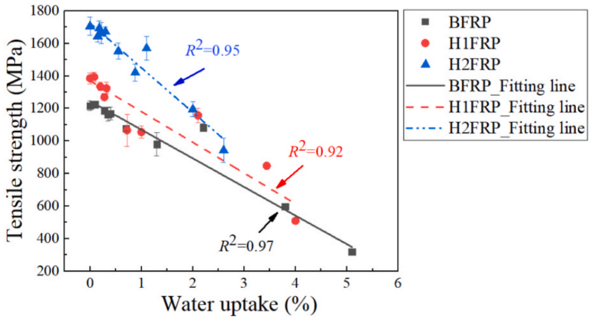


Fig. 11. Fitting curves for TS and water uptake data.

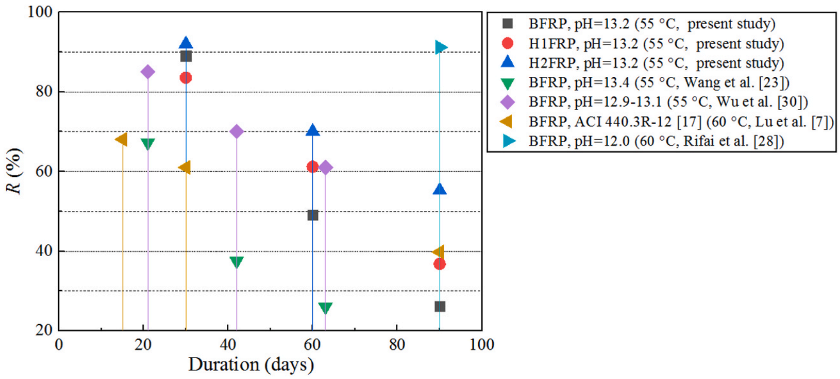


Fig. 12. Residual tensile strength data obtained in the present study and previous studies conducted in similar environments.

50–60 °C because the effects of resin plasticisation on the bars are diminished.

Yi et al. [5] obtained strong evidence of the importance of alkalinity in the deterioration of BFRP bars. Generally, a high-alkalinity solution with high OH^- concentrations leads to a high FRP degradation rate [29]. Therefore, as shown in Fig. 12, the *R* value of the BFRP bars examined herein (pH = 13.2) was slightly higher than that recorded by Wang et al. (pH = 13.4) [23] but lower than those reported by Wu et al. (pH = 12.9–13.1) [30] and Rifai et al. (pH = 12.0) [28]. Additionally, after 90 d of conditioning, the TSs of the H1FRPs and H2FRPs were significantly higher than those of the BFRPs and the samples prepared by Lu et al. [7] (ACI 440.3R-12 [17]), in particular, H2FRP bars. These results suggest that the long-term degradation rate of the HFRP bars decreased with increasing hybrid-carbon-fibre content in the bars. Similar results have also been reported for normal concrete- and SWSC-wrapped BFRP and HFRP bars conditioned in a seawater environment [14].

3.2.2. Load–displacement curves (LDCs)

The LDCs of the HFRP bars are shown in Figs. 13–15. Linear elastic behaviour was observed in all the LDCs of the unconditioned and conditioned BFRP and H2FRP bars. However, a yield platform with yield plastic deformation was observed in the LDCs of the H1FRP bars when ~80% of the ultimate tensile load (P_u) was achieved. The yield platform for H1FRP bar cannot be readily

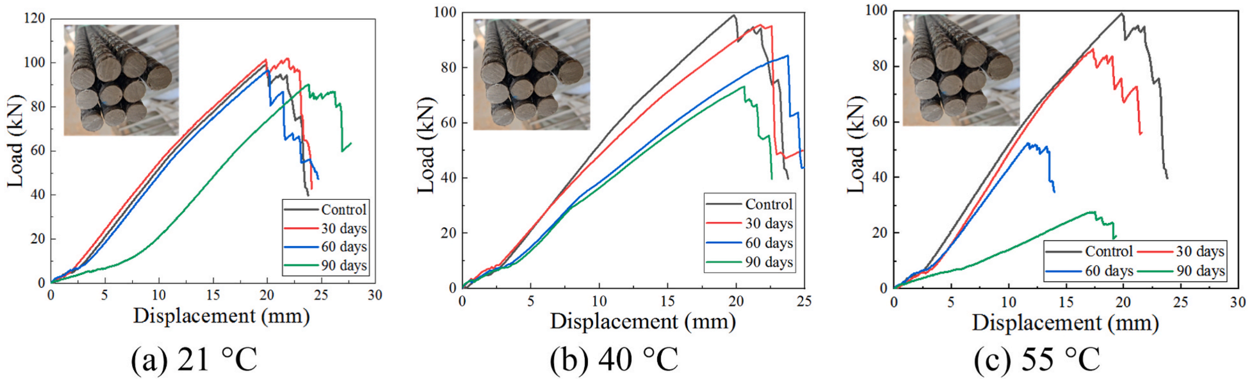


Fig. 13. LDCs of the BFRP bars.

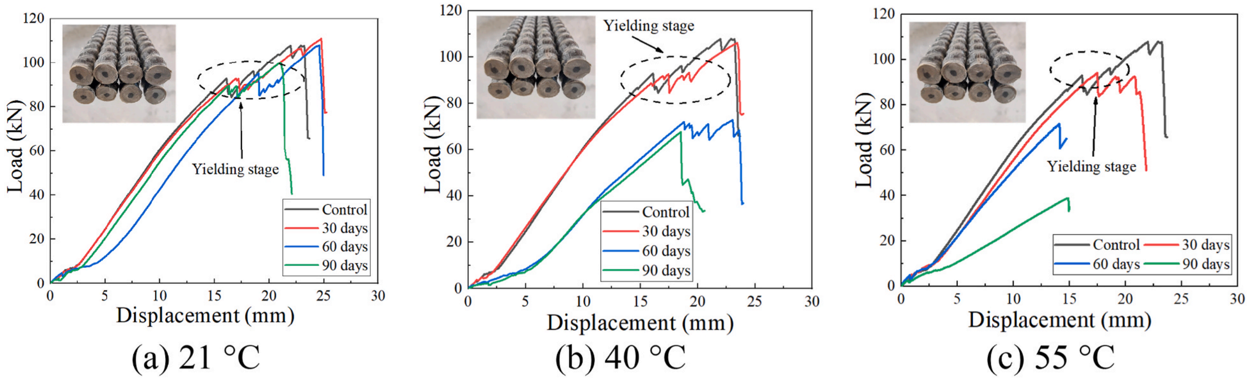


Fig. 14. LDCs of the H1FRP bars.

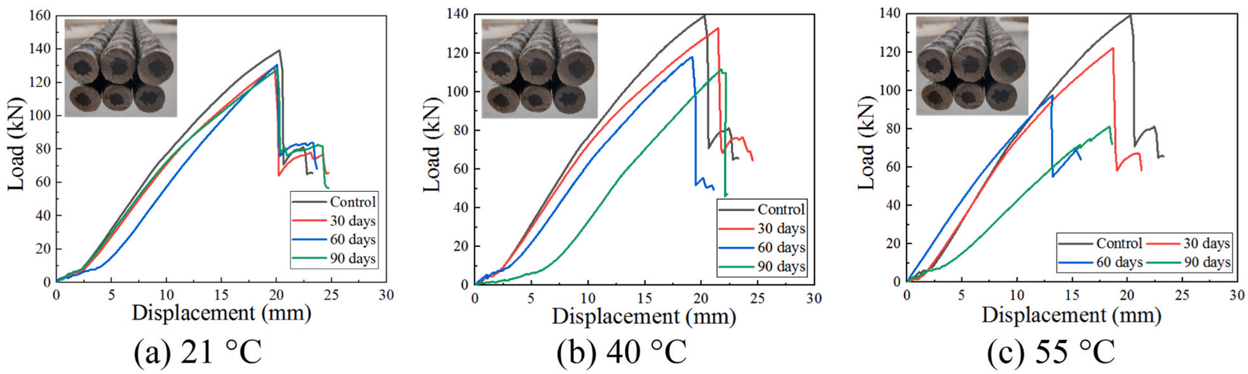


Fig. 15. LDCs of the H2FRP bars.

distinguished between the yield and strengthening stages; however, this is particularly important for brittle BFRP bars.

According to Li et al. [31], the tensile load (P) of BFRP bars has a relatively short-term decline with oscillations as it approaches the ultimate load (P_{u-b}), and the subsequent failure is accompanied by longitudinal delamination caused by the interlaminar stress. Moreover, the P_u value of FRP bars is believed to increase with increasing diameter [32]. In the LDCs of unconditioned BFRP and H1FRP bars (Fig. 16), an interesting observation could be made in terms of the load corresponding to the first appearance of the yield platform for the H1FRP bars, which was ~0.9 P_{u-b}. A basalt-fibre:carbon-fibre ratio of 9:1 was adopted for the H1FRP specimens in this study. The P_u of laminated plates is similar to the sum of the $\sum P_{u-i}$ of single-layer plates, which leads to a clear platform section in the stress–strain curve of a laminated plate, owing to the existence of interlaminar shear stress [33]. Therefore, the yield platform in the LDC of the H1FRP bars represents the failure process of the external BFRP and internal CFRP, respectively.

In contrast, a yield platform was not observed in the LDCs of the H2FRP bars, which also indicates the significance of the carbon-

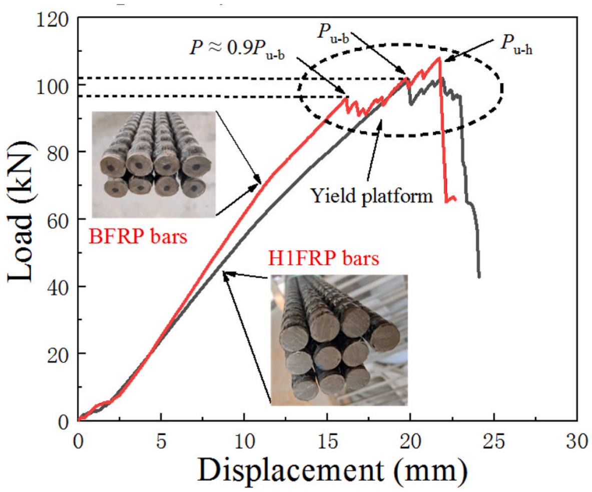


Fig. 16. LDCs of unconditioned BFRP and H1FRPbars.

fibre hybridisation content in the failure modes of HFRP bars. Additionally, it is worth noting that the yield platform of the H1FRP bars vanished with corrosion, possibly because the corroded BFRP in the outer layer continued to be restrained and supported by the CFRP in the inner layer after failure, which was caused by the lag of shear-force transfer owing to the decent interlayer bonding strength [33].

3.3. Morphological analysis

3.3.1. Surface morphology

Fig. 17 shows surface morphologies of the conditioned HFRP specimens, which reveal minor changes at 21 °C, and more yellowed ribs with a fuzzy and coarse surface texture at 40 °C. In contrast, the samples conditioned at 55 °C showed the most notable changes, as a brighter and more porous surface than that of the reference samples was observed, with an exposed solution that was muddy, dark yellow. These results indicate depolymerisation of the resin and its consequent dissolution in the simulated solution [5,23]. Similar surface morphologies were observed on the HFRP bars owing to their outer layers being circumferentially wrapped by BFRP.

3.3.2. Microstructural morphology

The microstructural morphologies of the surfaces and interfaces of the unconditioned and conditioned HFRP bars examined by DM are shown in Fig. 18. For the reference FRP samples, resin-rich areas were attached to the fibre surface and filled the gap between fibres, indicating suitable adhesion between the resin and fibres and at their interface.

DM analysis of the corroded HFRP specimens revealed gaps between the fibres and resin, along with bare fibres and fragmented resin on the surface (Fig. 18 (a)) and interface (Fig. 18 (b) and (c)), suggesting that all specimens had similar deterioration mechanisms that were affected by temperature. The damage to the fibres, resin, and fibre–resin interface impeded the transfer of stress between the fibres, and consequently decreased the TS [32]. Nevertheless, compared with BFRP bars, the deterioration mechanism of the H1FRP and H2FRP bars had several unique features based on their microstructural morphologies, as summarised below.

(1) Degradation primarily occurred in the outer BFRP layer, whereas that of CFRP mostly involved resin hydrolysis and a certain extent of fibre–resin interfacial debonding.
(2) The degree of BFRP degradation near the BFRP–CFRP interface increased, and a greater degree of resin hydrolysis occurred in the BFRP specimens near the BFRP–CFRP interface.
(3) A greater extent of degradation such as resin hydrolysis occurred at the BFRP–CFRP interfaces in the H2FRP bars than that in the H1FRP bars.

These results indicate that increasing the carbon-fibre content in the HFRP bars mitigated resin hydrolysis and fibre–resin interfacial debonding. Similar results have been obtained by Guo et al. [14].

3.4. FTIR and NMR analyses

HFRP bar samples were comprehensively characterised to elucidate the deterioration mechanism of their resin matrix. The FTIR spectra of reference and corroded FRPs (90 d at 55 °C) are shown in Fig. 19, along with those of the BFRP bars for comparison. Changes in the functional groups of the resin matrix during exposure can lead to variations in the corresponding FTIR band in the spectra, as indicated by Manalo et al. [9].

After conditioning, the characteristic ester peak at 1730 cm^{-1} essentially disappeared and the OH^{-} band enhanced in intensity, indicating hydrolysis of the resin [18]. Additionally, a new band was observed at 840 cm^{-1}, which was attributed to the changes in the vibration characteristics of the ether group caused by water invasion of the sample via hydrogen bond formation with the C–O–C group

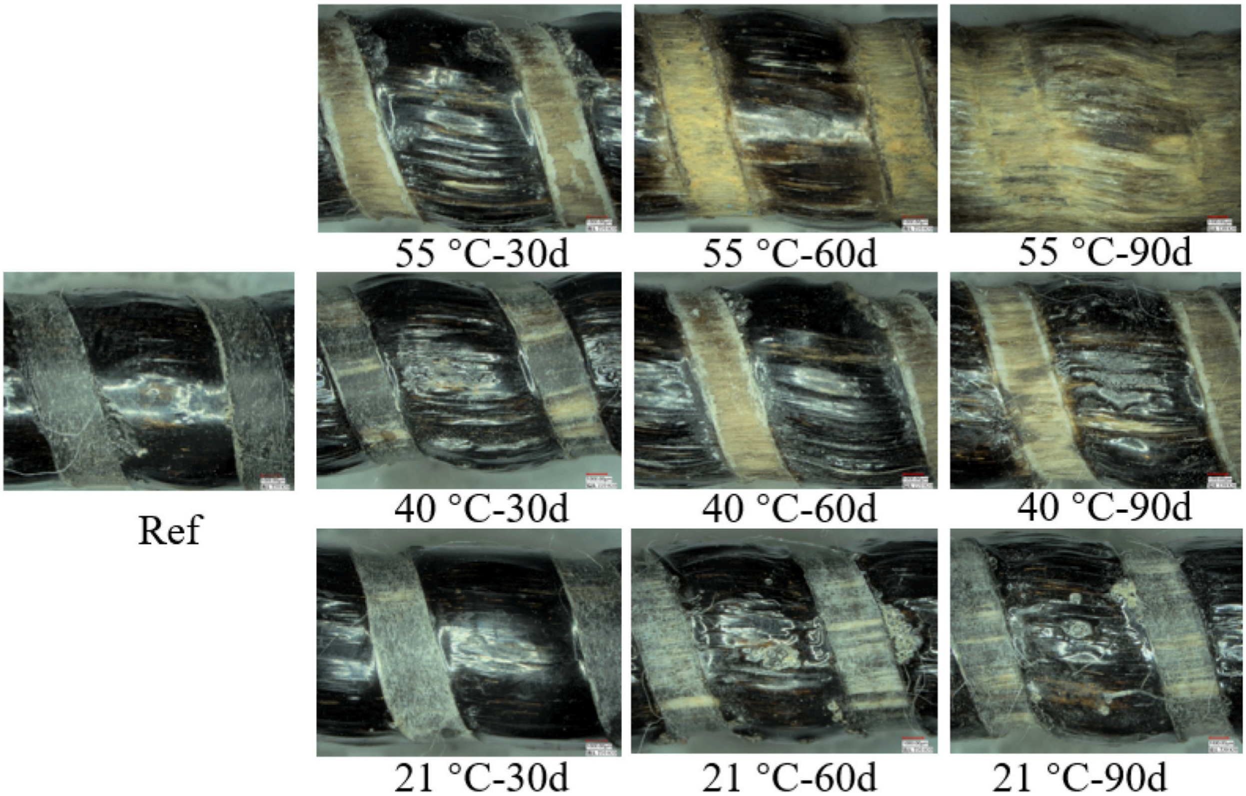


Fig. 17. Surface morphologies of HFRP bars.

[22]. The changes in the C–OH vibration band due to hydration led to hydrogen bond formation (C–O–H···OH_2), yielding a characteristic peak at ~1125 cm^{-1} [34]. These results suggest that the matrix degradation was associated with changes in the chemical structure.

^{1}H and ^{13}C NMR spectroscopies were further conducted to analyse the structures of the dissolved products (Fig. 20), which confirmed the hydrolysis of the ester in the epoxy. Typically, alkaline solutions are conducive to resin hydrolysis. The carbon–oxygen and hydrogen–oxygen single bonds in the ester and water, respectively, are ruptured, and a hydroxyl group is added at the R′–O– end (R and R′ represent hydrocarbon groups) [18]. This implies that the hydrolysis of the epoxy in alkaline solutions significantly influences the deterioration of HFRP bars. The plausible reaction mechanism is illustrated in Fig. 21.

3.5. *Matrix digestion analysis*

The detailed results of the matrix digestion tests are listed in Table 8. The M_t values of the reference BFRP, H1FRP, and H2FRP specimens were estimated to be 27.86%, 26.29%, and 25.56%, respectively (Fig. 22 (a)), which indicated that the M_t values decreased with increasing carbon-fibre volume fraction in the HFRP bars, possibly because of the low matrix content in the CFRP (Fig. 10). Fig. 22 (b) shows the M_t values of the conditioned specimens, in which those of the HFRPs decreased with increasing temperature. The M_t values of the BFRP, H1FRP, and H2FRP bars decreased from 21.44%, 18.59%, and 21.25% (40 °C) to 11.23%, 15.56%, and 18.55% (55 °C), respectively. Similar findings on BFRPs and GFRPs have been reported by Rifai et al. [28] and El-Hassan et al. [35], respectively. Notably, the H2FRP bars exhibited the highest M_t values, whereas the BFRPs showed the lowest M_t values, except at 40 °C.

3.6. *Fibre etching*

Fibre degradation was not observed in the conditioned HFRP bars, as noted in the DM-based morphological analysis. Scheffler et al. [36] and Liu et al. [37] reported that the corrosion features of the glass–basalt-fibre system included pitting and corrosion shells. Furthermore, Guo et al. [34] indicated that these features can be observed when fibres are conditioning in alkaline solutions at room temperature for several days or even hours. Additionally, the loss in strength of FRP bars in alkaline solutions, as determined using short-beam shear tests, is primarily due to the fibre–resin interfacial debonding and resin hydrolysis [22,29,38]. However, the reduction in tensile properties—a fibre-dominated behaviour—is primarily caused by fibre degradation. Therefore, the elements of etched fibres should be determined to clarify the deterioration mechanism of the HFRP bars.

Fig. 23 shows the concentrations of Si, Al, and Fe that leached from the fibres in the HFRP bars after conditioning. It is worth noting that all bars were placed in a solution-containing vessel at identical temperatures. Therefore, the concentration of the elements

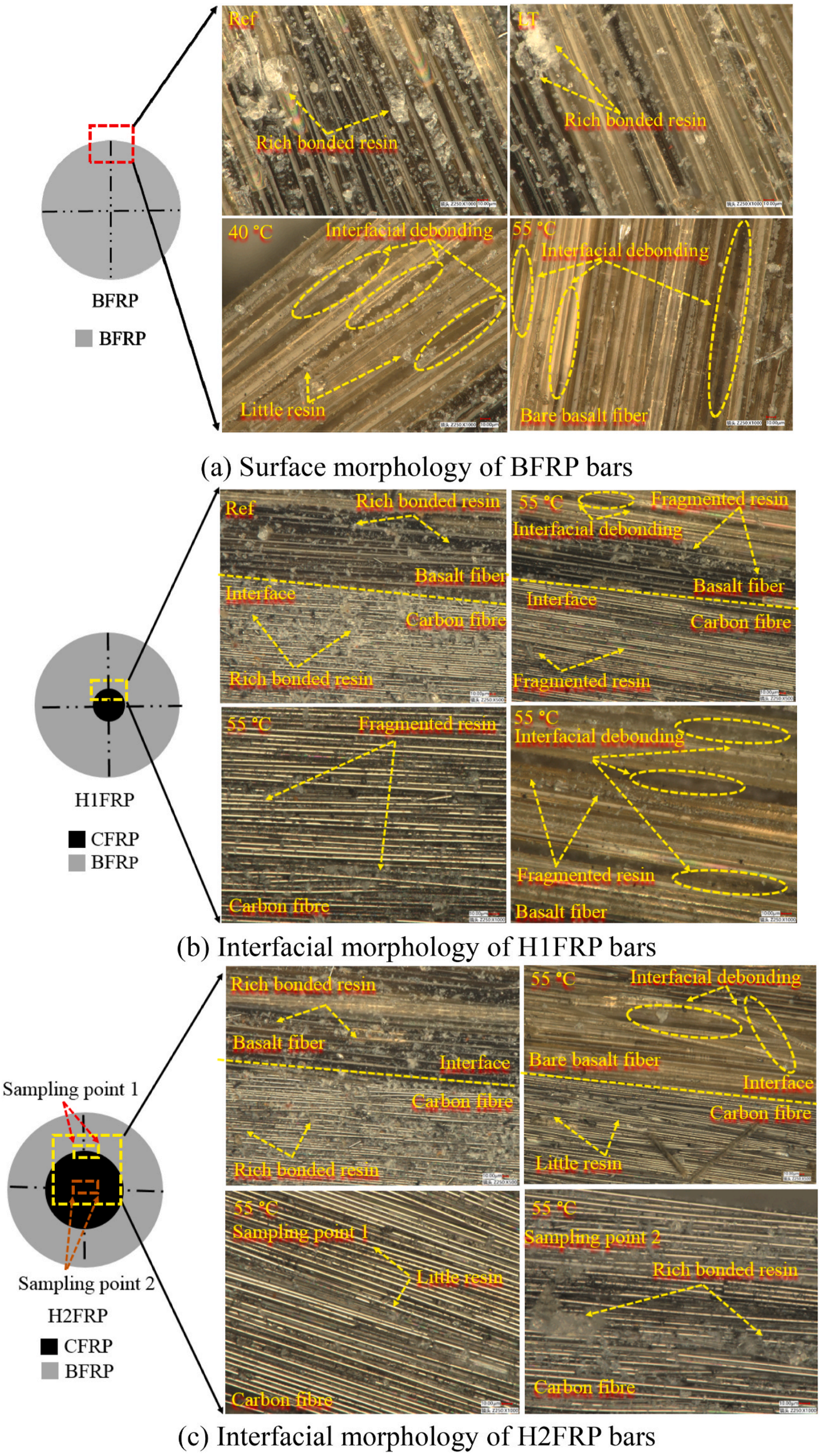
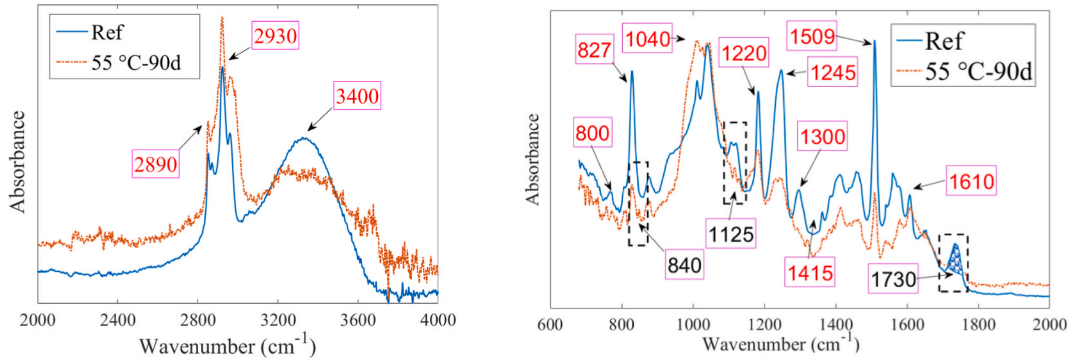


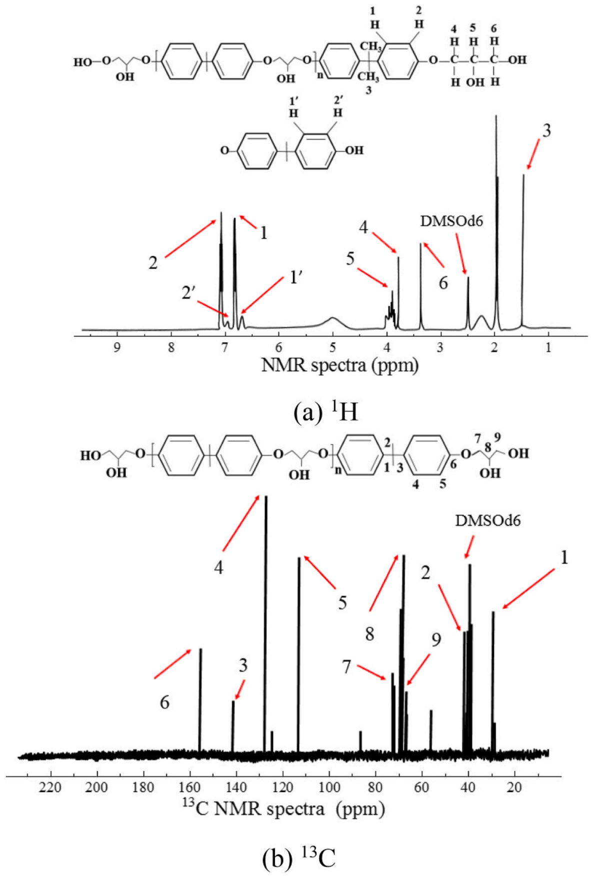
Fig. 18. Microstructural morphologies of the HFRP bars.

corresponded to the cumulative average concentration of the specimens. The Al, Fe, and Si contents significantly increased with increasing temperature. These results indicate deterioration of the fibres, in particular, basalt fibres. Generally, the OH^- ions in alkaline solutions are conducive to fibre degradation and disintegration of their $-Si-O-Si$ and $-Al-O-Si$ cross bonds [23]. Zhu et al. [39] suggested that aluminium oxide can be dissolved by OH^- in alkaline solutions (pH > 9), as follows:



(a) 2000–4000 cm^{-1} (b) 600–2000 cm^{-1}

Fig. 19. Microstructural morphologies of the FRP bars.



(b) ^{13}C

Fig. 20. NMR analysis of the resin matrix.

$$Al_2O_3 + 2OH^-(adsorbed) \rightarrow 2AlO_2^-(\text{aqueous}) + H_2O \tag{5}$$

Moreover, the Cl^- in simulated SWSC solutions can promote this reaction [40].

$$Al^{3+}_{(\text{in crystal lattice of the oxide})} + 2Cl^- + 2OH^- \rightarrow Al(OH)_2^- \tag{6}$$

Additionally, the detection of Fe by ICP-MS implied the occurrence of the following reactions:

$$Fe^{2+} + Cl^- \rightarrow FeCl_{complex} \tag{8}$$

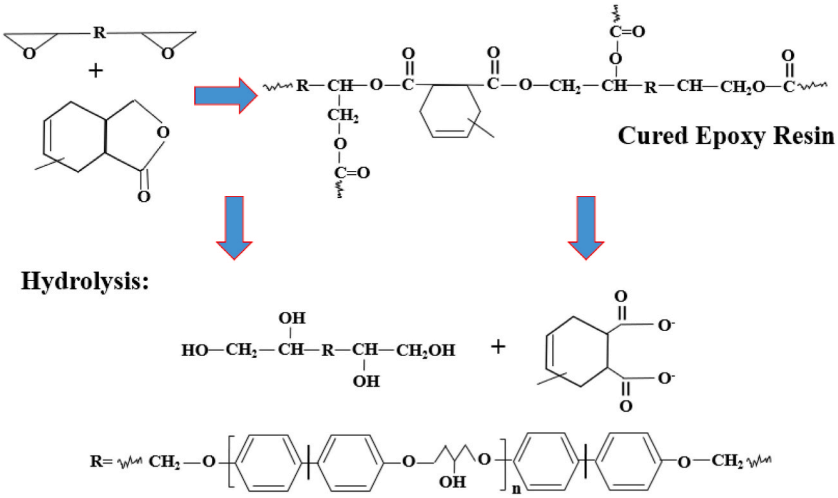


Fig. 21. Plausible reaction mechanism of ester hydrolysis.

Table 8
Matrix contents of HFRP bars.

FRP bar specimen	Conditioning		Content (%)	
	Environment (°C)	Duration (d)	Fibre	Matrix
BFRP	Ref.	/	72.14	27.86
	21	90	72.69	27.31
	40		78.56	21.44
	55		88.77	11.23
H1FRP	Ref.	/	73.71	26.29
	21	90	74.59	22.78
	40		81.41	18.59
	55		84.33	15.56
H2FRP	Ref.	/	74.44	25.56
	21	90	74.92	22.08
	40		78.75	21.25
	55		81.45	18.55

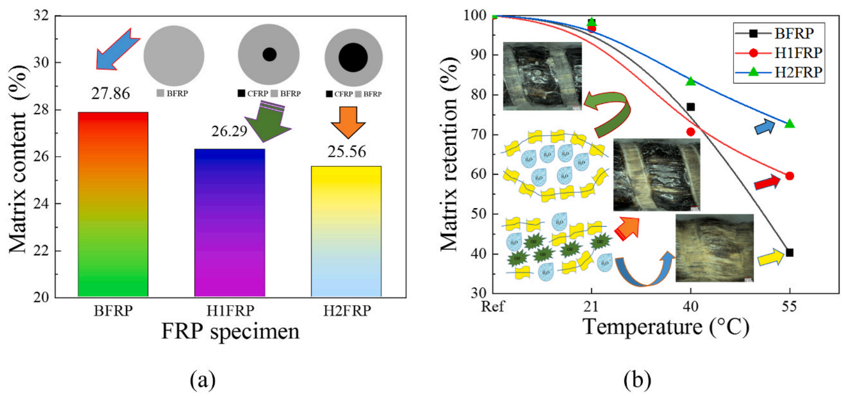


Fig. 22. Matrix (a) content and (b) retention in conditioned and unconditioned HFRP bars.

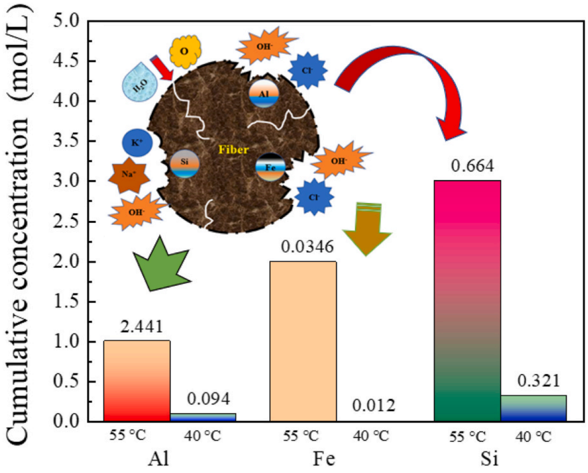


Fig. 23. Concentrations of leached elements of fibres in HFRP bars.

$$[FeCl_{complex}]^{-} + OH^{-} \rightarrow Fe(OH)_2 + Cl^{-} \tag{9}$$

$$Fe(OH)_2 + O_2 + H_2O \rightarrow Fe(OH)_3 \rightarrow Fe_2O_3 \bullet nH_2O \tag{10}$$

Furthermore, Wu et al. [30] and Lu et al. [41] included the alkali–silica reaction in the degradation of BFRP bars in SWSC environments, as follows:

$$Na^{2+} + (K^{+}) + SiO_2 + OH^{-} \rightarrow Na(K) - Si - H_{\rm gel} \tag{11}$$

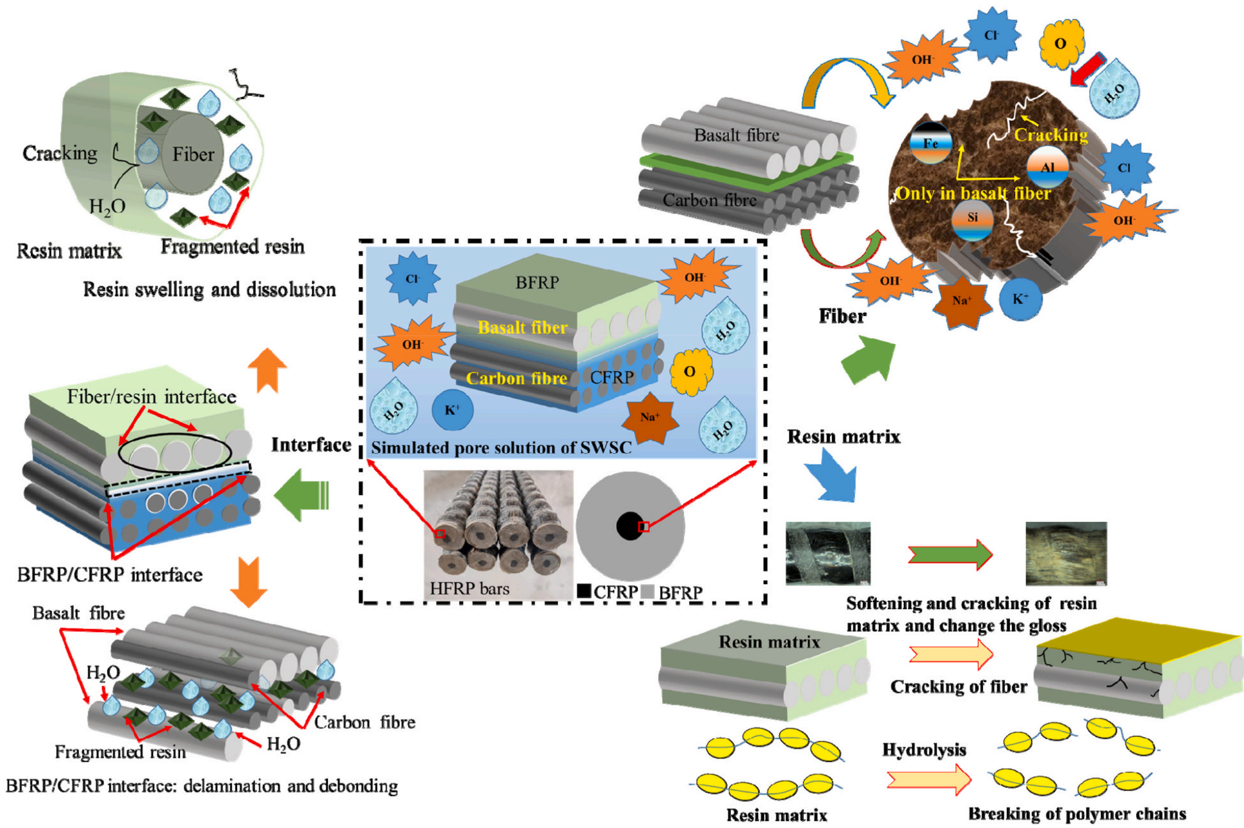


Fig. 24. Deterioration mechanism of HFRP bars in a simulated SWSC pore solution.

The osmotic coefficient of alkaline water-based solutions, such as NaCl solutions, increases with increasing molal concentration of $FeCl_2$ [42]. Therefore, the iron-oxide content considerably influences the water uptake of HFRPs, which also explains the higher saturated water absorption of BFRP bars.

3.7. Discussion

The overall evaluation of the deterioration mechanism of HFRP bars conditioned in a simulated SWSC pore solution is illustrated in Fig. 24. Deterioration was initiated on the outer layer of the BFRP, as the conditioning duration increased. First, the HFRP samples absorbed moisture (Fig. 7), the water molecules permeating through resin and fibre–resin interface exhibited a plasticisation effect and induced hydrolysis (Figs. 17–22) and swelling reactions, which led to fibre–resin interfacial delamination and debonding (Fig. 18). Moreover, the hydrolysis and plasticisation of the resin presumably caused fibre exposure or resin cracking. The OH^-, K^+, Na^+, and Cl^- ions in the exposed solution reacted with the Si, Al, and Fe in the fibres, resulting in fibre etching and surface cracks (Fig. 23), which, in turn, possibly affected the resin and fibre–resin interface. This deterioration was accelerated at high temperatures; therefore, the softening of the resin led to a gradual change in the surface gloss of the bars (Fig. 17). Additionally, the effective moisture barrier in the annular basalt–carbon-fibre interfacial area (Fig. 10) caused local deterioration of BFRP in the HFRP bars, but delayed their overall performance degradation.

In summary, moisture caused the initial deterioration of the resin and initiated the deterioration of the HFRP bars. The corrosive ions in simulated SWSC water-based solutions at high temperatures aggravated this deterioration. Nonetheless, increasing the carbon-fibre content in HFRP improved the mechanical strength of the bars and considerably alleviated the deterioration of their tensile properties, which is critical for realising high-bearing-capacity, durable FRP–SWSC structures. Furthermore, the yield platform of the H1FRP bars with a carbon-fibre content of 10% was particularly important for brittle FRP bars, as it provided potential visual and direct pre-emptive signals of failure of the FRP–SWSC structures. However, further investigations on the durability of SWSC-wrapped basalt–carbon-fibre HFRP bars in terms of field exposure can provide further insight into these signals.

4. Conclusions

The effects of carbon-fibre content on the durability of HFRP bars in a simulated SWSC pore solution were comparatively evaluated. The tensile strengths of different HFRP bars in various solution-based environments were determined. Water uptake, DM, FTIR, NMR, matrix digestion, and ICP-MS analyses were performed to assess the degradation. The following conclusions can be drawn:

(1) Increasing the carbon-fibre content in the HFRP bars from 0% to 25% clearly enhanced their mechanical properties, such as tensile strength and elastic modulus, (1216.18–1705.99 MPa and 61.17–85.01 GPa, respectively).
(2) Water uptake of the HFRP bars was strongly dependent on the carbon-fibre-hybrid content and conditioning temperature, with higher water absorption and diffusion rates occurring at higher conditioning temperatures. Under similar exposure conditions, the water uptake rate decreased with increasing hybrid-carbon-fibre content in the bars. Increasing the carbon-fibre volume fraction in HFRP from 0% to 25% decreased the saturated water absorption and diffusivity coefficient from 5.18% to 3.49% and 5.04×10^{-13} to 3.98×10^{-13} $m^2\ s^{-1}$, respectively.
(3) The TS of the HFRP bars decreased with increasing conditioning temperature and duration. However, the TS degradation rate was overall alleviated with increasing carbon-fibre content in the HFRP bars. After 90 d of conditioning at 55 °C, the BFRP, H1FRP, and H2FRP bars retained 26.2%, 36.78%, and 55.26% of their initial TSs, respectively.
(4) The carbon-fibre content also played an important role in determining the failure mode of the HFRP bars. A yield platform with yield plastic deformation was observed in the LDCs of the H1FRP bars with a carbon-fibre content of 10% when ~80% of the ultimate tensile load was achieved. However, this signal was not observed in the LDCs of the other types of HFRP bars with carbon-fibre contents of 0% and 25%.
(5) The degradation of the HFRP bars was primarily caused by resin hydrolysis, fibre etching, and interfacial debonding and delamination, in which the resin deteriorated via water absorption, thereby initiating deterioration of the HFRP bars. Moreover, the annular basalt–carbon-fibre interfacial region in the HFRP bars was a weak component, which should be focused on in future research on durability.

Author statement

Xiangke Guo: Methodology, Writing - original draft, Data curation. Chuangsheng Xiong: Conceptualization, Writing review & editing, Funding acquisition. Zuquan Jin: Conceptualization, Writing review & editing, Project administration, Funding acquisition. Tao Sun: Supervision, Formal analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This study is a part of a series of projects financially supported by the National Natural Science Foundation of China (U1806225, 52078259, and U2106221), the Key Research and Development Plan of Shandong Province (No. 2019GGX102053), the MOUNT TAI

Scholarship of Shandong Province (ts20190942), the Open Research Fund of Key Laboratory of Construction and Safety of Water Engineering of the Ministry of Water Resources (China Institute of Water Resources and Hydropower Research; No. IWHR-ENGI-202010), and the Ningbo 2025 Science and Technology Major Project (No. 2020Z035). Additionally, this study was also supported by the National 111 project.

References

[1] M. Bazli, M. Heitzmann, B.V. Hernandez, Hybrid fibre reinforced polymer and seawater sea sand concrete structures: a systematic review on short-term and long-term structural performance, Construct. Build. Mater. 301 (2021), 124335.
[2] Z. Jin, C. Xiong, T. Zhao, Y. Du, X. Zhang, N. Li, Y. Yu, P. Wang, Passivation and depassivation properties of Cr–Mo alloyed corrosion-resistant steel in simulated concrete pore solution, Cement Concr. Compos. 126 (2022), 104375.
[3] Z.Q. Jin, X. Zhao, Y.J. Du, S.Y. Yang, D.Q. Wang, T.J. Zhao, Y. Bai, Comprehensive properties of passive film formed in simulated pore solution of alkali-activated concrete, Construct. Build. Mater. 319 (2022), 126142.
[4] J. Xiao, C. Qiang, A. Nanni, K. Zhang, Use of sea-sand and seawater in concrete construction: current status and future opportunities, Construct. Build. Mater. 155 (2017) 1101–1111.
[5] Y. Yi, S. Guo, S. Li, M. Zillur Rahman, L. Zhou, C. Shi, D. Zhu, Effect of alkalinity on the shear performance degradation of basalt fiber-reinforced polymer bars in simulated seawater sea sand concrete environment, Construct. Build. Mater. 299 (2021), 123957.
[6] C.H. Lu, M.Z. Ni, T.S. Chu, L.Y. He, Comparative investigation on tensile performance of FRP bars after exposure to water, seawater, and alkaline solutions, J. Mater. Civ. Eng. 32 (2020), 04020170.
[7] Z. Lu, Y. Li, J. Xie, Durability of BFRP bars wrapped in seawater sea sand concrete, Compos. Struct. 255 (2021), 112935.
[8] T. D'Antino, M.A. Pisani, Influence of sustained stress on the durability of glass FRP reinforcing bars, Construct. Build. Mater. 187 (2018) 474–486.
[9] A. Manalo, G. Maranan, B. Benmokrane, P. Cousin, O. Alajarmeh, W. Ferdous, R. Liang, G. Hota, Comparative durability of GFRP composite reinforcing bars in concrete and in simulated concrete environments, Cement Concr. Compos. 109 (2020), 103564.
[10] Y. Pan, D. Yan, Study on the Durability of GFRP Bars and Carbon/glass Hybrid Fiber Reinforced Polymer (HFRP) Bars Aged in Alkaline Solution, Composite Structures, 2020, 113285.
[11] Y.I. Tsai, E.J. Bosze, E. Barjasteh, S.R. Nutt, Influence of hygrothermal environment on thermal and mechanical properties of carbon fiber/fiberglass hybrid composites, Compos. Sci. Technol. 69 (2009) 432–437.
[12] C. Li, G. Xian, H. Li, Combined effects of temperature, hydraulic pressure and salty concentration on the water uptake and mechanical properties of a carbon/glass fibers hybrid rod in salty solutions, Polym. Test. 76 (2019) 19–32.
[13] Y. Swolfs, L. Gorbatikh, I. Verpoest, Fibre hybridisation in polymer composites: a review, Compos. Appl. Sci. Manuf. 67 (2014) 181–200.
[14] X. Guo, Z. Jin, C. Xiong, T. Sun, N. Li, Y. Yu, X. Zhang, Deterioration of mechanical properties of basalt/carbon hybrid FRP bars in SWSC under seawater corrosive environment, Construct. Build. Mater. 317 (2022), 125979.
[15] L.C. Bank, T.R. Gentry, B.P. Thompson, J.S. Russell, A model specification for FRP composites for civil engineering structures, Construct. Build. Mater. 17 (2003) 405–437.
[16] ASTM Standard ASTM D570, Water Absorption of Plastics. ASTM D570-18, ASTM International, West Conshohocken, Philadelphia, Pa, 2018, 19103.
[17] ACI 440.3R-12. Guide Test Methods Forfiber-Reinforced Polymers (FRPs) for Reinforcing or Strengthening Concrete Structures, ACI Committee, 2012.
[18] P. Yang, Q. Zhou, X.-X. Yuan, J. Kasteren, Y.-Z. Wang, Highly efficient solvolysis of epoxy resin using poly(ethylene glycol)/NaOH systems, Polym. Degrad. Stabil. 97 (2012) 1101–1106.
[19] ASTM, Standard Test Methods for Constituent Content of Composite Materials, D3171, ASTM, West Conshohocken, PA, 2015.
[20] T. Liu, X. Liu, P. Feng, A comprehensive review on mechanical properties of pultruded FRP composites subjected to long-term environmental effects, Compos. B Eng. 191 (2020), 107958.
[21] L.-R. Bao, A.F. Yee, Effect of temperature on moisture absorption in a bismaleimide resin and its carbon fiber composites, Polymer 43 (2002) 3987–3997.
[22] Z.K. Wang, X.L. Zhao, G.J. Xian, G. Wu, R.K.S. Raman, S. Al-Saadi, Durability study on interlaminar shear behaviour of basalt-, glass- and carbon-fibre reinforced polymer (B/G/CFRP) bars in seawater sea sand concrete environment, Construct. Build. Mater. 156 (2017) 985–1004.
[23] Z.K. Wang, X.L. Zhao, G.J. Xian, G. Wu, R.K.S. Raman, S. Al-Saadi, A. Haque, Long-term durability of basalt- and glass-fibre reinforced polymer (BFRP/GFRP) bars in seawater and sea sand concrete environment, Construct. Build. Mater. 139 (2017) 467–489.
[24] American Society of Civil Engineers (ASCE), Pre-standard for Load and Resistance Factor Design (LRFD) of Pultruded Fiber Reinforced Polymer, FRP) structures, 2010.
[25] EUR27666 L. Ascione, J.F. Caron, P. Godonou, K.V. IJselmuijden, J. Knippers, T. Mottram, M. Oppe, Gantriis, M.G. Sorensen, J. Taby, L. Tromp, Prospect for New Guidance in the Design of FRP, Publications Office of the European Union, 2016.
[26] China Association for Engineering Construction Standardization (CECS), Technical Specification for Pultruded Fiber Reinforced Polymer Composites Structure, 2019.
[27] P. Feng, Y. Wu, T. Liu, Non-uniform fiber-resin distributions of pultruded GFRP profiles, Compos. B Eng. (2021), 109543.
[28] M. Al Rifai, H. El-Hassan, T. El Maaddawy, F. Abed, Durability of basalt FRP reinforcing bars in alkaline solution and moist concrete environments, Construct. Build. Mater. 243 (2020), 118258.
[29] Z. Lu, L. Su, S. Tan, Y. Li, J. Xie, F. Liu, Long-term shear performance of bare and cement mortar-coated BFRP bars in corrosive environments, Construct. Build. Mater. 237 (2020), 117658.
[30] G. Wu, Z.Q. Dong, X. Wang, Y. Zhu, Z.S. Wu, Prediction of long-term performance and durability of BFRP bars under the combined effect of sustained load and corrosive solutions, J. Compos. Construct. 19 (2015), 04014058.
[31] Y. Li, S. Yin, Y. Lu, C. Hu, Experimental investigation of the mechanical properties of BFRP bars in coral concrete under high temperature and humidity, Construct. Build. Mater. 259 (2020), 120591.
[32] A.H. Ali, H.M. Mohamed, B. Benmokrane, Bar size effect on long-term durability of sand-coated basalt-FRP composite bars, Compos. B Eng. 195 (2020), 108059.
[33] H.T. Hahn, S.W. Tsai, On the behavior of composite laminates after initial failures, J. Compos. Mater. 8 (3) (1974) 288–305.
[34] F. Guo, S. Al-Saadi, R.K.S. Raman, X.L. Zhao, Durability of fiber reinforced polymer (FRP) in simulated seawater sea sand concrete (SWSSC) environment, Corrosion Sci. 141 (2018) 1–13.
[35] H. El-Hassan, T. El-Maaddawy, A. Al-Sallamin, A. Al-Saidy, Performance evaluation and microstructural characterization of GFRP bars in seawater-contaminated concrete, Construct. Build. Mater. 147 (2017) 66–78.
[36] C. Scheffler, T. Förster, E. Mäder, G. Heinrich, S. Hempel, V. Mechtcherine, Aging of alkali-resistant glass and basalt fibers in alkaline solutions: evaluation of the failure stress by Weibull distribution function, J. Non-Cryst. Solids 355 (2009) 2588–2595.
[37] M.-Y. Liu, H.-G. Zhu, N.A. Siddiqui, C.K.Y. Leung, J.-K. Kim, Glass fibers with clay nanocomposite coating: improved barrier resistance in alkaline environment, Compos. Appl. Sci. Manuf. 42 (2011) 2051–2059.
[38] H.-Y. Kim, Y.-H. Park, Y.-J. You, C.-K. Moon, Short-term durability test for GFRP rods under various environmental conditions, Compos. Struct. 83 (2008) 37–47.
[39] D. Zhu, W.J. van Ooij, Corrosion protection of AA 2024-T3 by bis-[3-(triethoxysilyl)propyl]tetrasulfide in neutral sodium chloride solution. Part 1: corrosion of AA 2024-T3, Corrosion Sci. 45 (2003) 2163–2175.

[40] E.-S. Sherif, Corrosion and corrosion inhibition of aluminum in arabian gulf seawater and sodium chloride solutions by 3 amino-5-mercapto-1,2,4-triazole, Int. J. Electrochem. Sci. 6 (2011) 1479–1492.
[41] Z. Lu, S. Tan, P. Huang, Z. Lei, F. Liu, J. Xie, Durability of cement mortar-covered BFRP bars in simulated seawater environment, Construct. Build. Mater. 234 (2020), 117803.
[42] R. Lemire, U. Berner, C. Musikas, D. Palmer, P. Taylor, O. Tochiyama, Chemical Thermodynamics of Iron, Part 1, OECD/Nuclear Energy Agency, Thermodynamic Data Bank, 2013.